Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/06/2013
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Objective Monitoring of Crew Neurobehavioral Functions in Mars 520-Day Simulation		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/01/2010	**End Date:**	02/28/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date change to 2/28/2013 per NSBRI (Ed., 2/26/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Metaxas, Dimitris (Rutgers University) Mollicone, Daniel (Pulsar Informatics, Inc.) Basner, Mathias (University of Pennsylvania)		
Grant/Contract No.:	NCC 9-58-NBPF00006		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Astronauts/Cosmonauts on long-duration missions will endure the isolation and confinement of space to a greater degree than previous travelers. In order to be able to carry out critical asks during such missions, crews must maintain a high level of neurobehavioral and psychosocial performance fitness in the face of prolonged isolation, demanding workload and work schedules, and other stressors. The Russian Institute for Biomedical Problems (IBMP) 520-day simulated mission to Mars (in a special simulated spacecraft chamber located at IBMP in Moscow) offers a unique and critically-needed opportunity for evaluating key aspects of crewmembers' neurobehavioral functions (i.e., mood, cognitive performance), sleep and wake activity patterns, and psychosocial conflicts while confined and socially isolated for a period of time longer than the 180-day missions currently occurring on ISS, and longer than the longest spaceflight mission conducted to date by any country. This project uses novel, minimally obtrusive technologies to reliably detect stress, fatigue, performance changes, negative emotional states, and conflicts in the 6 male crewmembers participating in IBMP's 520-day simulated Mars mission. Our technologies were tested for feasibility in IBMP's 105-day pilot study in the same simulated spacecraft chamber being used in the Mars 520-day study. Lessons learned from the 105-day pilot project helped to improve the reliability of our assessments for the 520-day mission. This project used three state-of-the-art objective technologies—wrist actigraphy, the PVT Self Test and optical computer recognition (OCR) of facial expressions of fatigue, stress and emotions—that we have developed and validated in laboratory and analog studies. Both the PVT Self test and OCR videos are being obtained on crewmembers using modified notebook computers, with 10-minute data samples obtained twice a day (AM and PM), once each week throughout the mission, to keep the assessments minimally obtrusive while maintaining the ability to acquire data throughout the 520-day mission—as stated above, the priority aims of the study focus on the unique feature of the Mars 520 analog, namely the 520-day mission duration. The video collection for OCR analysis occurred during the 3-min. PVT Self Test. In addition to PVT Self Test and OCR facial video, our weekly test sessions of crewmembers include subjective assessments of sleep quality, workload, conflicts, and mood states, using brief validated visual analog scales (VAS), the Profile of Mood States Short Form (POMS-SF) and Beck Depression Inventory (BDI-II), and a novel conflict-tension frequency scale. The 520-day mission began with hatch closing on June 3, 2010 (June 4 was mission day 1). A combination of a NASA/NSBRI Liaison at IBMP and an engineering company (Pulsar Informatics, Inc.) have greatly facilitated data acquisition from the technologies we deployed in the IBMP chamber, while redundant technology (i.e., backups) have facilitated high quality data. Data acquisition has been very successful to date using the procedures we developed based on lessons learned from the 105-day pilot study in the chamber. The six crewmembers completed all (100%) baseline (pre-mission) data acquisitions. Data acquisition was completed November 4, 2011. Data acquisition rates were >98.0% complete (73,203 hours of data) for actigraphy of sleep, wake and rest states, as well as light exposure, and 100% complete (888 test bouts) for subjective sleep reports and PVT-B performance tests throughout the 520-day period of mission confinement.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The 3-minute PVT Self Test used in the Mars 520-day study is being developed to help people quickly and objectively detect the extent to which fatigue is affecting their neurobehavioral alertness (i.e., tendency to be impulsive or to lapse) and psychomotor speed. As such, the technology has high potential for usefulness in a range of safety-sensitive environments on Earth. Potentially any occupation in which alertness and fatigue management (or fitness for duty testing) are essential to prevent errors on critical tasks will benefit from adaptations of the PVT Self Test technology (e.g., airport security screeners, night shifts workers, first responders, etc.). The automated optical computer recognition (OCR) technology being developed for objective monitoring and management of stress, negative emotional states and fatigue in space flight has applications for many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
Task Progress:	Data acquisition was completed November 4, 2011. Data acquisition rates were >98.0% complete (73,203 hours of data) for actigraphy throughout the 520-day study and 100% complete (888 test bouts) for subjective sleep reports and PVT-B performance tests. Primary data analyses were completed and published: Basner, M., Dinges, D.F., Mollicone, D.J., Ecker, A.J., Jones, C.W., Hyder, E., Di Antonio, A., Savelev, I., Kan, K., Goel, N., Morukov, B.V., Sutton, J.P. Mars 520-d mission simulation reveals protracted crew hypokinesis and alterations of sleep duration and timing. Proceedings of The National Academy of Sciences of The USA. 2013 Feb 12;110(7):2635-40. A Second paper has been submitted for publication: Basner, M., Dinges, D.F., Mollicone, D.J., Savelev, I., Ecker, A.J., Di Antonio, A., Jones, C.W., Hyder, E., Kan, K., Morukov, B.V., Sutton, J.P. Behavioral and Psychosocial Changes during a 520-Day Simulated Interplanetary Mission to Mars.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Basner M, Di Antonio A, Ecker AJ, Mollicone DJ, Dinges DF. "Sleep-Wake Patterns and Psychomotor Vigilance Performance During a 520-Day Simulated Mission to Mars." 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Basner M, Di Antonio A, Jones C, Ecker AJ, Kan K, Mollicone DJ, Dinges DF. "Sleep-Wake Behavior and PVT Performance During a 520-day Simulated Mission to Mars." 26th Annual Meeting of the Associated Professional Sleep Societies, Boston, MA, June 9-13, 2012. Sleep. 2012;35 Suppl:A78. , Jun-2012
Abstracts for Journals and Proceedings	Basner M, Dinges DF, Mollicone DJ, Savelev I, Ecker AJ, Di Antonio A, Jones CW, Hyder EC, Kan K, Morukov B, Sutton JP. "Behavioral and Psychosocial Changes During a 520-Day Simulated Interplanetary Mission to Mars." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013

Abstracts for Journals and Proceedings	Basner M, Dinges DF, Mollicone DJ, Savelev I, Ecker AJ, Di Antonio A, Jones CW, Hyder EC, Kan K, Morukov B, Sutton JP. "Behavioral and Psychosocial Changes During A 520-Day Simulated Interplanetary Mission to Mars." 19th IAA Humans in Space Symposium, Cologne, Germany, July 7-13, 2013. 19th IAA Humans in Space Symposium, Cologne, Germany, July 7-13, 2013. Abstract #137. , Jul-2013
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Articles in Peer-reviewed Journals	Basner M, Dinges DF, Mollicone D, Ecker A, Jones CW, Hyder EC, Di Antonio A, Savelev I, Kan K, Goel N, Morukov BV, Sutton JP. "Mars 520-d mission simulation reveals protracted crew hypokinesis and alterations of sleep duration and timing." Proceedings of the National Academy of Sciences of the United States of America. Proc Natl Acad Sci U S A. 2013 Feb 12;110(7):2635-40. http://dx.doi.org/10.1073/pnas.1212646110 ; PubMed PMID: 23297197 , Feb-2013
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